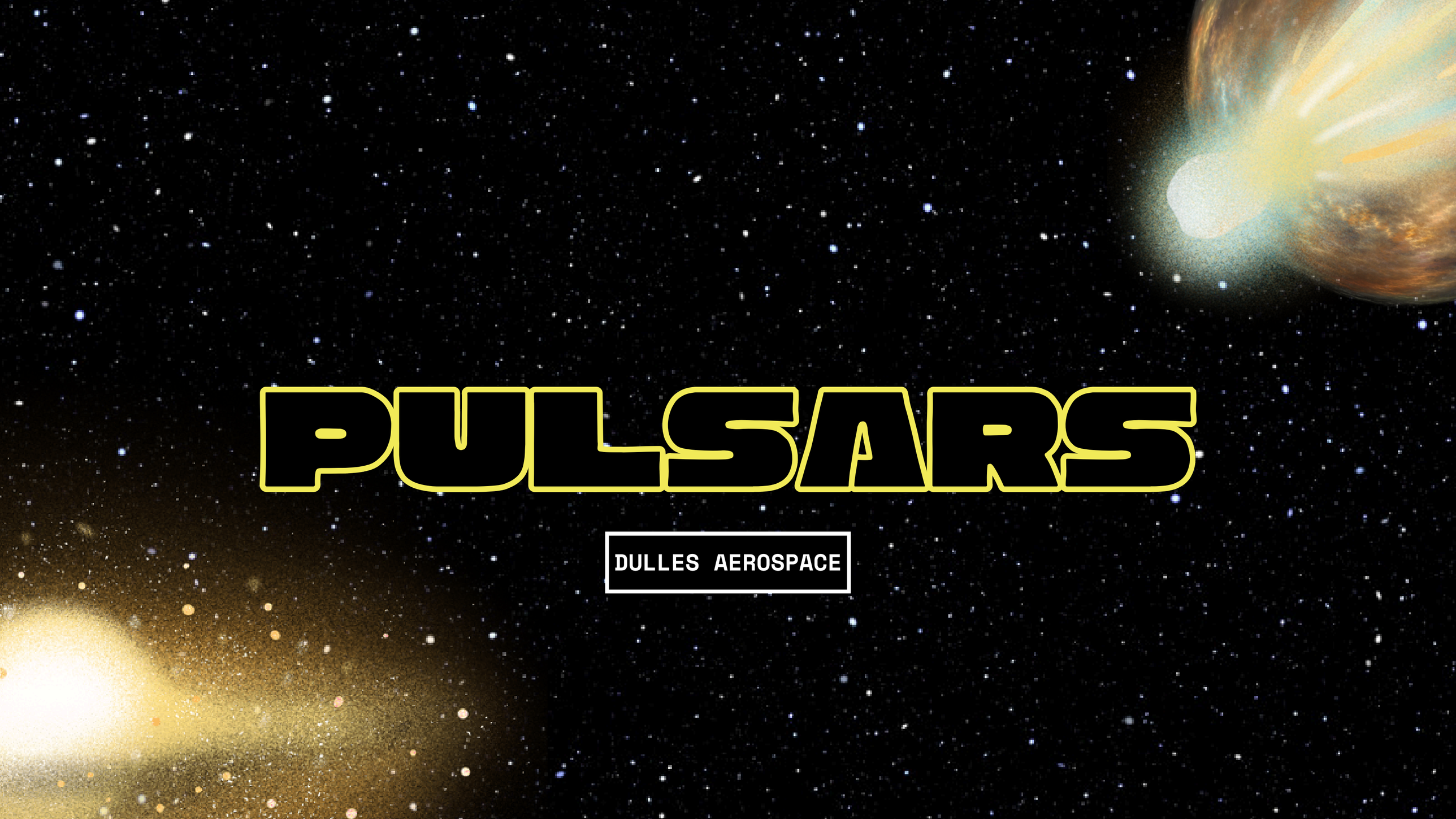

PULSARS
DULLES AEROSPACE

NOTES

DO YOU WANT SHIRTS? WE CAN HAVE A TEAM DESIGN COMPETITION.

1. OFFICER APPS DUE OCT 1
2. FILL OUT THE DRONE FORM TODAY.
3. ATTENDANCE

MAKE SURE EVERYONE IS IN SLACK/REMIND

NOTE: NEXT WEEK PLS BRING PERSONAL LAPTOPS (OR ATLEAST UR PHONE) WE WILL BE SETTING UP ACCOUNTS FOR THE RESEARCH

TODAY YOURE LEARNING STUFF.. THERE WILL BE A BLOOKET.. MAYBE THERE ARE PRIZES. THIS INFO WILL BE REALLY CRUCIAL FOR THE RESEARCH YOURE DOING LATER!

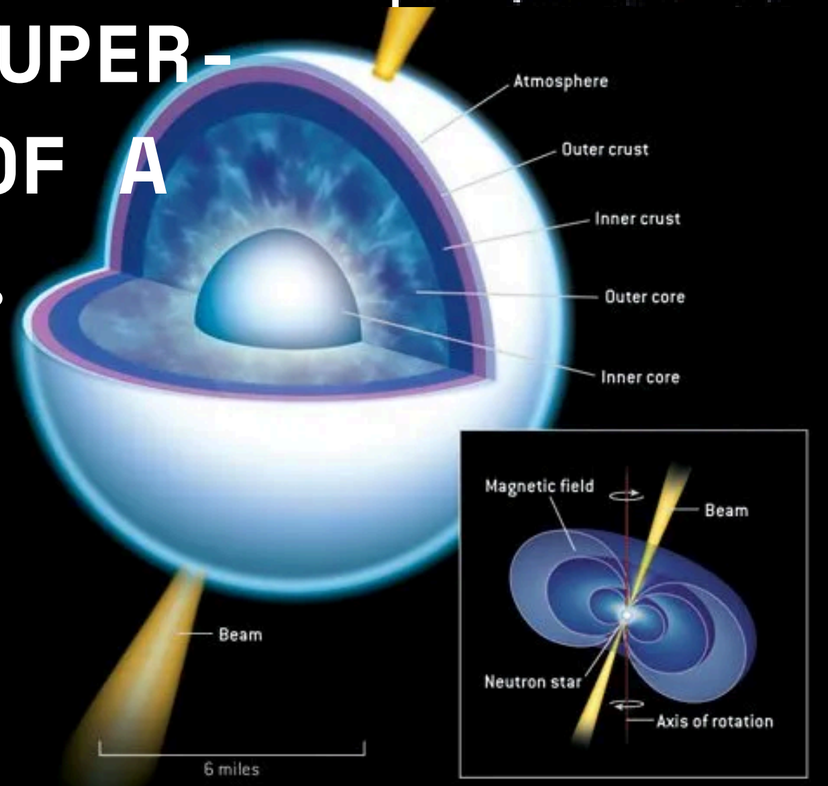

ARE YOU
READY?

WHAT IS A PULSAR?

A PULSAR IS A RAPIDLY SPINNING, HIGHLY MAGNETIZED NEUTRON STAR THAT EMITS REGULAR BEAMS OF ELECTROMAGNETIC RADIATION

WHATS A NEUTRON STAR

A NEUTRON STAR IS THE SUPER-DENSE COLLAPSED CORE OF A MASSIVE GIANT STAR.



WHY DO PULSARS "PULSE"

THEY DONT

SO WHAT DO THEY DO

THE ANSWER IS A COMBINATION OF CONSERVATION OF ANGULAR MOMENTUM, FLUX CONSERVATION, AND RELATIVISTIC ELECTRODYNAMICS. (MAKE SURE TO ASK QUESTIONS AT THE END OF EACH SLIDE SINCE IT ALL BUILDS. USE THE WHITEBOARD)

FIRST...

PULSARS SPIN DUE TO CONSERVATION OF ANGULAR MOMENTUM. (L=IW).

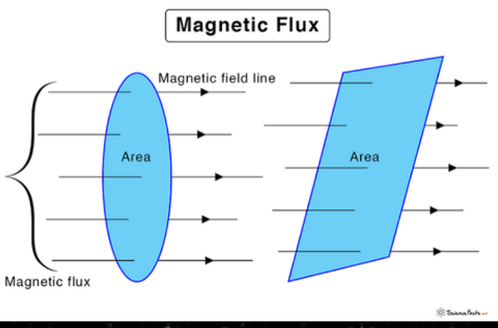
THINK..

WHEN A MASSIVE STAR ($M>=8M_{\odot}$) EXHAUSTS NUCLEAR FUEL.. CORE UNDERGOES GRAVITATIONAL COLLAPSE. SOME THINGS HAPPEN.

1. **MOMENT OF INERTIA DROPS:** $I = 2/5\ MR^2$ FOR A SPHERE. DURING COLLAPSE, R DROPS FROM ABOUT 10^6 KM TO ABOUT 10 KM
2. SINCE R SCALES DOWN BY A FACTOR OF 10^5 I DECREASES BY A FACTOR OF 10^{10} . TO KEEP L CONSTANT, ANGULAR VELOCITY INCREASES BY FACTOR OF 10 BILLION!!!! (**VERY FAST**)

A STAR THAT ORIGINALLY TOOK MAYBE A MONTH TO ROTATE NOW ROTATES HUNDREDS OF TIMES PER SECOND!!! (VERY FAST)

SECOND....



JUST LIKE ANGULAR MOMENTUMS CONVERVED... SO IS THE STARS TOTAL **MAGNETIC FLUX!**.

$$\Phi = \int \mathbf{B} \cdot d\mathbf{A}$$

ITS CONSERVED WITHIN THE HIGHLY CONDUCTIVE PLASMA DURING THE COLLAPSE. (IF YOU REALLY WANT TO KNOW WHY SEARCH UP ALFVENS THEOREM FOR IDEAL PLASMAS. BUT WE DONT HAVE TIME TO GO INTO THAT)

1. **AMPLIFIED FIELD STRENGTH:** SURFACE AREA DROPS BY A FACTOR OF 10^{10} AND MAGNETIC FIELD STRENGTH HAS TO SCALE UP BY 10^{5} **(SEE ABOVE FORMULA)**. A NORMAL STELLAR CORE FIELD OF 100 GAUSS SCALES UP TO A HYPER-INTENSE PULSAR FIELD OF 10^{12} GAUSS.

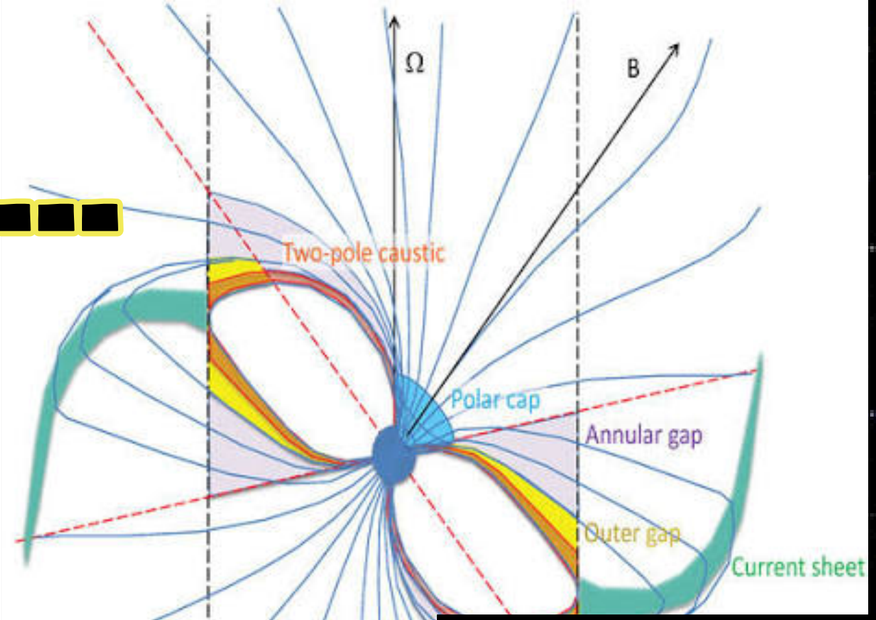
SECOND (CONTINUED)....

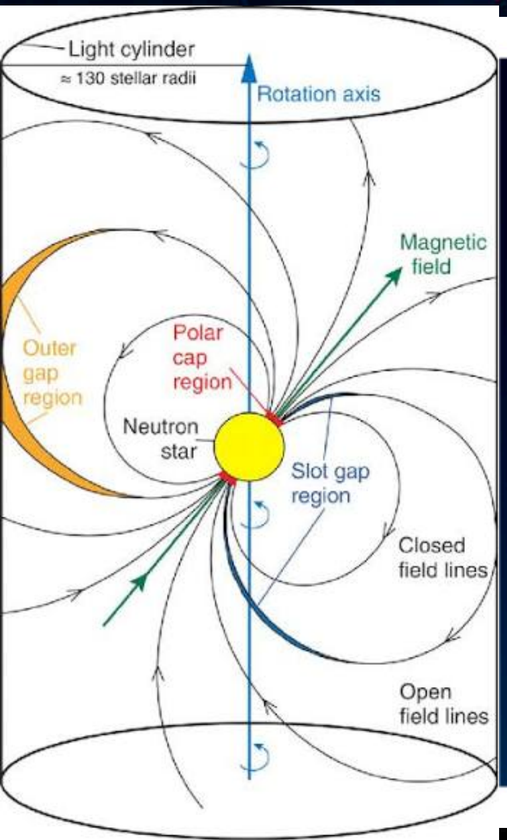
1. **AMPLIFIED FIELD STRENGTH!**
2. **INDUCED ELECTRIC FIELD:** THE MASSIVE DIPOLE SPINS RAPIDLY, CREATING A SUPER BIG INDUCED ELECTRIC FIELD (E=-V X B). THE VOLTAGE DIFFERENCE BETWEEN THE PULSARS EQUATOR AND POLES CAN REACH 10^{16} VOLTS (A LOT)
3. **MAGNETOSPHERE:** THE ELECTRIC FIELD EXERTS ELECTROSTATIC FORCE ON ELECTRONS/IONS THAT IS WAYYY STRONGER THAN THE STARS GRAVITY. IT RIPS THOE CHARGED PARTCILES STRAIGHT OUT OF THE CRYSTALINE IRON CRUST.: FILLING THE SURROUNDING SPACE WITHA DENSE PLASMA CALLED... THE **MAGNETOSPHERE!!**

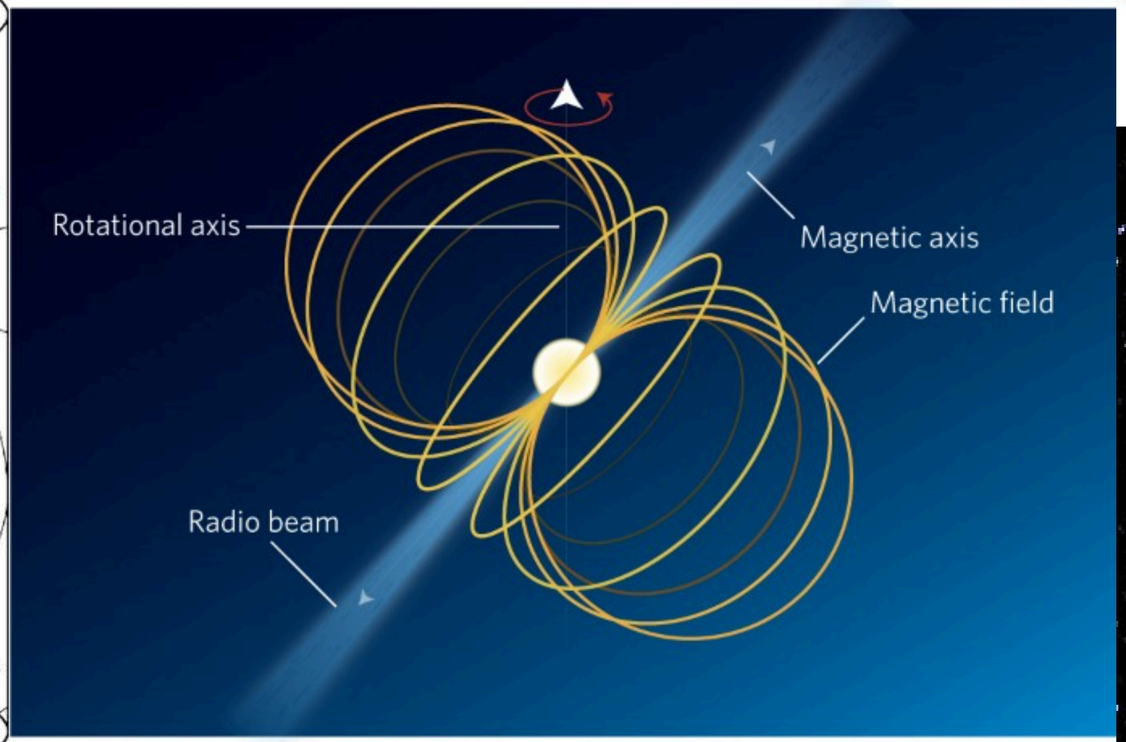
I.I KNOW THAT MAY HAVE BEEN CONFUSING. TO SUMMARIZE.: IT SPINS SUPER FAST AND HAS A SUPER BIG MAGNETIC FIELD. THE MAGNETIC FIELD INDUCES AN ELECTRIC FIELD CREATING A POTENTIAL DIFFERENCE. THAT POTENTIAL FIELD EXERTS ENOUGH FORCE TO RIP IONS OUT OF THE STARS GRAVITY CREATING A DENSE PLASMA MAGNETOSPHERE.

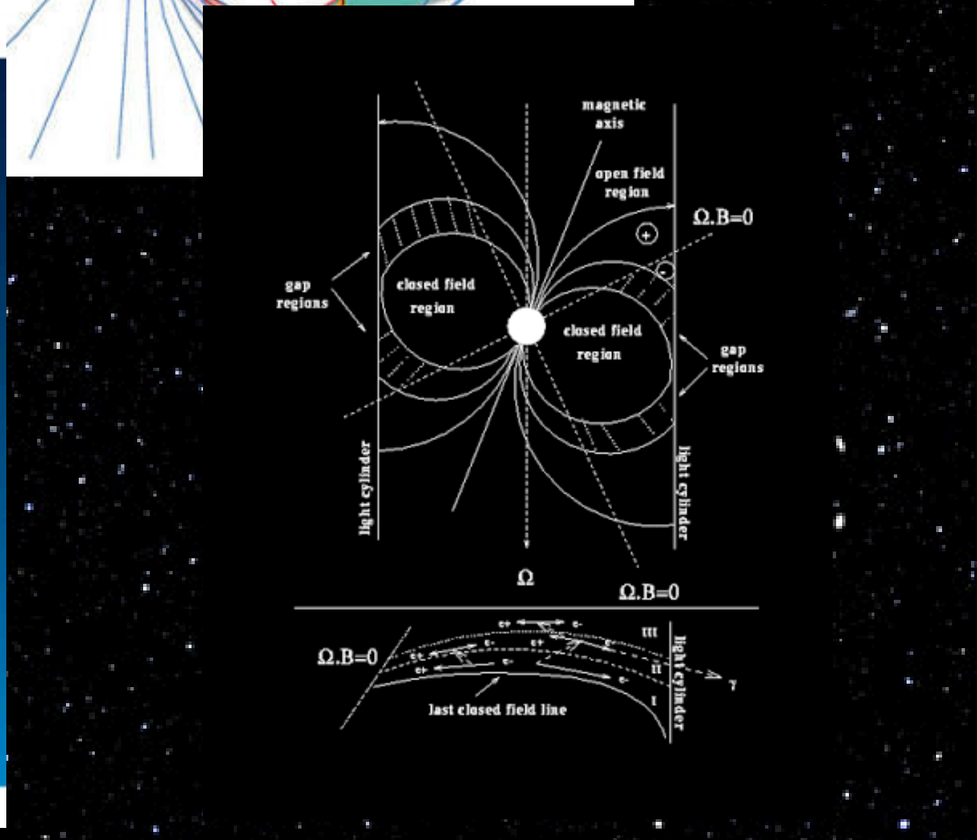
SECOND (CONTINUED)....









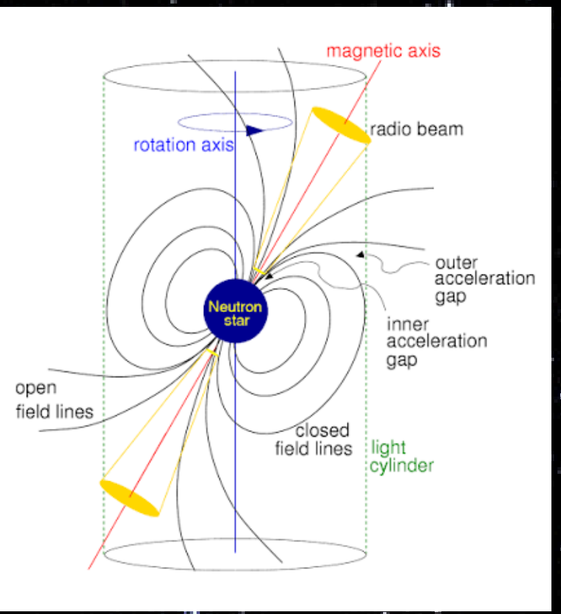


THIRD...

THE PARTICLE ACCELATOR (NOT AS COMPLEX AS IT SOUNDS)

THE MAGNETOSPHERE IS FORCED TO CO-ROTATE WITHTHE NEUTRON STAR LIKE “SPOKES” ON A WHEEL. BUT THAT CANT EXTEND INFINITELY. AS A FAR ENOUGH DISTANCE, THE SPEED NEEDED TO KEEP UP WITH THE STARS ROTATION REACES THE SPEED OF LIGHT. (CALLED THE LIGHT CYLINDER DEFINED BY THE RADIUS):

$$R_{LC} = C/W$$



1. **OPEN VS CLOSED FIELD LINES:** MAGNETIC FIELD LINES THAT LOOP BACK TO THE STAR **BEFORE** REACHING R_LC REMAIN CLOSED AND TRAP THAT PLASMA. FIELD LINES THAT WOULD CROSS THE LIGHT CYLINDER CAN’T CLOSE SINCE THE PLASMA WOULD NEED TO TRAVEL FASTER THAN LIGHT. SO... THEY OPEN INTO SPACE!

THIRD (CONTINUED)...

1. **OPEN VS CLOSED FIELD LINES**
2. **CURVATURE RADIATION:** ALONG THESE OPEN FIELD LINES AT THE MAGNETIC POLES, ELECTRONS/POSITRONS ARE ACCELETRATED AT.. (WELL SUPER FAST SPEEDS TO SUMMARIZE). AS THEY ARE FORCED TO FOLLOW THE CURVED MAGNETIC FIELD LINES.. THEY EMIT HIGH ENERGY PHOTONS VIA CURVATURE RADIATION:
   A. **OVERSIMPLIFYING A LOT:** CURVATURE RADIATION IS WHEN A CHARGED PARTICLE MOVING AT ALMSOT THE SPEED OF LIGHT HAS TO TURN A CORNER - THAT MEANS IT HAS TO ACCELERATE (DIRECTION WISE). SO IT SHEDS KINETIC ENERGY AS ELECTROMAGNETIC RADIATION. BC OF HOW FAST IT IS, THE LIGHT GETS COMPRESSED INTO A VERY TIGHT BEAM DIRECTLY INFRONT OF THE PARTICLES PATH.

A place where this process is thought to be important is at the poles of rotating neutron stars (pulsars):

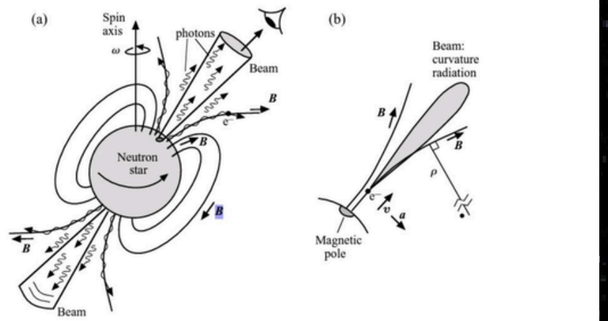


Fig. 8.12: (a) Spinning neutron star with offset magnetic pole. Electrons spiral outward along magnetic field lines and beam radiation in the polar direction. (The spiraling electrons illustrated would not radiate into the beam.) (b) Magnetic pole showing the radius of curvature ρ and curvature-radiation beam of a single emerging electron.

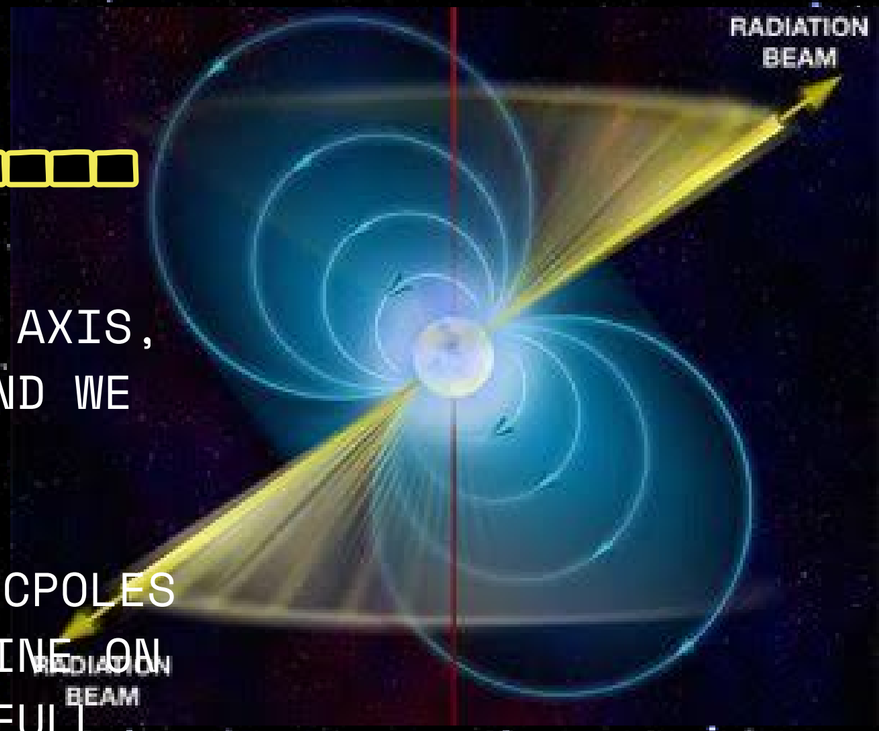
THIRD (CONTINUED)...

1. **OPEN VS CLOSED FIELD LINES**
2. **CURVATURE RADIATION:** (I KNOW THAT PROBABLY DIDNT REALLY MAKE SENSE. ITS FINE. JUST REMEMBER THE TURNING = ACCELERATION = SHED ENERGY = RELEASE ELCTROMAGNETIC RADITATION)
3. **PAIR PRODUCTION CASCADES:** THE EMITTED HIGH-ENERGY GAMMA-RAY PHOTONS INTERACT WITH THE INTENSE MAGNETIC FIELD TO SPONTANEOUSLY SPLIT INTO ELECTRON-POSITRON PAIRS. THEY ACCELERATE AND EMIT MORE GAMMA RAYS TRIGGERING A RUNAWAY CASCADE.
4. **COHERENCE:** THE PLASMA BUNCHES INTO MACRO PARTICLES THAT MOVE TOGETHER (BASICALLY THINK ABOUT IT LIKE A LASER OF RADAITION, GENERATING SUPER BRIGHT BEAMS OF RADIO WAVES, X-RAYS, AND GAMMA RAYS OUT OF THE MAGNETIC POLES)

THIRD (CONTINUED)...

1. SO BASICALLY.. SOME MAGNETIC FIELD LINES DONT CLOSE. THE PARTICLES IN THOSE FIELD LINES HAVE TO FOLLOW THE PATH OF THEM.. WHICH IS CURVED. THAT MEANS THEY HAVE TO DIRECTIONALLY ACCELERATE AND SO THEY SHED KINETIC ENERGY EMITTING GAMMA RAYS. THE EMITTED GAMMA RAYS UNDERGO THE SAME THING, SPLITTING AND THEN EMITTING MORE.. THAT ENERGY BUNDLES TOGETHER AND SHOOTS OUT OF THE POLES LIKE A SUPER HIGH POWER LASER.
2. (ITS OKAY IF THE PHYSICS OF IT ISNT INTUITVE - THIS IS JUST BACKGROUND INFO.. AS LONG AS YOU VAGUELY GET IT YOURE GOOD)

FOURTH.....



THE PULSE... THE MISALIGNTED DIPOLE.
IF THE MAGNETIC AXIS WERE PERFECTLY ALIGNED WITH THE SPIN AXIS, THE BEAMS WOULD **CONSTANTLY** POINT IN THE SAME DIRECTION AND WE WOULD JUST SEE A STEADY SOURCE OF LIGHT.
BUT... ITS ALMOST **ALWAYS** AT A TILTED ANGLE.
AS THE NEUTRON STAR SPINS ON ITS ROTATION AXIS, THE MAGNTICPOLES SWEEP THROUGH SPACE IN A WIDE CONE. IF EARTH HAPPENS TO LINE ON THE PATH OF THIS CONE,, WE CAN DETECT THE BEAM ONCE PER FULL ROTATION WHEN EARTH INTERSECTS THE BEAM. THE “PULSE” WE OBSERVE ISNT THE STAR TURNING ON AND OFF.. ITS THE HIGHLY FOCUSED BEAM OF RELAVISTIC PARTICLE RADIATION SWEEPING ACROSS OUR LINE OF SIGHT.

SO.. ALL YOU NEED TO KNOW IS. IT DOESNT PULSE. IT EMITS ENERGY THROUGH THE MAGNETIC POLES (WHICH WE CAN DETECT) CONSTANTLY. THE PULSAR JUST SPINS WHICH AFFECTS WHEN WE CAN DETECT THAT RAY.
CREATING.. THE PULSE AFFECT.
THE TAKEAWAY FROM ALL THAT IS **PULSARS ARE LIKE LIGHTHOUSES.**

WE START RESEARCH

NEXT WEEK

WE'LL DO AN OVERVIEW OF THE PREREQUISTE COURSE AS WELL AS A LESSON ABOUT THE APPLICATION OF ALL THIS PHYSICS IN HOW WE ACTUALLY DETECT THINGS. THEN WE'LL SET UP YOUR ACCOUNTS SO YOU CAN DO STUFF.

DONT WORRY IF THIS SO FAR WAS KIND OF CONFUSING. YOU WONT USE THE PHYSICS ANYWAYS IN THE ACTUAL WORK BUT ITS GOOD TO (KIND OF) KNOW. THE MAIN POINT WAS THAT PULSARS = LIGHTHOUSES.

BLOOKET....

PULSARS CONT'D

DULLES AEROSPACE

NOTES...

1. OFFICER APPS DUE TODAY!!!!!!! BY 6PM.
2. FILL OUT THE DRONE FORM IF YOU HAVENT
3. ATTENDANCE

MAKE SURE EVERYONE IS IN SLACK/REMIND

NOTE: NEXT WEEK PLS BRING PERSONAL LAPTOPS (OR ATLEAST UR PHONE) WE WILL BE SETTING UP ACCOUNTS FOR THE RESEARCHING THING

SHIRTS.. DO WE WNAT THEM?

ALSO DUES..

ALSO TELL THEM ABT ASTEROIDS

OK GREAT

WE ESTABLISHED THAT PULSARS = LIGHTHOUSES.

OK SO IF PULSARS = LIGHTHOUSES...: HOW CAN WE USE THAT TO **TELL** IF WE'RE ACTUALLY LOOKING AT A PULSAR..

THATS THE TOPIC TODAY. (FOLLOWED BY RESEACH SET UP)

SO.

PULSARS ARE DETECTED BC THEYRE KIND OF LIKE ROTATING MAGNETIC BEAMS. WE SEE A “PULSE” EACH TIME THE BEAM SWEEPS ACROSS OUR LINE OF SIGHT.

BUT THEYRE PRETTY **FAR** AWAY. SWEEPING LIGHT YEARS OF SPACE WEAKENS THE BEAM. INDIVIDUAL LIGHTHOUSE SWEEPS END UP BEING HIDDEN BENEATH BACKGROUND NOISE.....

TO RECONSTRUCT FAINT BEAMS.. WE USE **PRESTO** TO STACK THOUSANDS OF SWEEPS TOGETHER INTO 4 CORE DIAGNOSTIC PLOTS.

WHAT WE START WITH IS RAW DATA FROM PARABOLIC DISHES FOCUSED ON EM RADIO WAVES ONTO A RECEIVER FOR 295-355 MHZ. THEN THOSE ARE AMPLIFIED, AND DIGITIZED. THEN THE INTENSITY OF THOSE BANDWIDTHS ARE RECORDED INTO A 3D ARRAY OF TIME, FREQUENCY, INTENSITY.

SO.

WELL RAW RADIO DATA CANT EXACTLY PRODUCE CLEAR PULSES... CAUSE INTERSTELLAR PLASMA SLOWS DOWN LOWER RADIO FREQUENCIES MORE THAN HIGHER ONES (DISPERSION).

WE HAVE TO DEDISPERSE - ARRIVAL TIMES ACROSS DIFFERENT FREQUENCY CHANNELS ARE SHIFTED IN TIME ACCORDING TO A TRIAL DISPERSION MEASURE (DM) TO ALIGN SIGNAL ACROSS ALL FREQUENCIES (LIKE JUST TESTING OUT UNTIL IT LOOKS UNSMEARED)

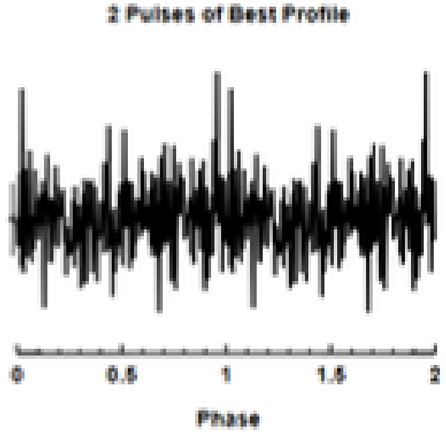
THEN THE PIPELINE DOES A FAST FOURIER TRANSFORM TO IDENTIFY REPEATING PERIODIC SPIKES IN NOISY SIGNALS

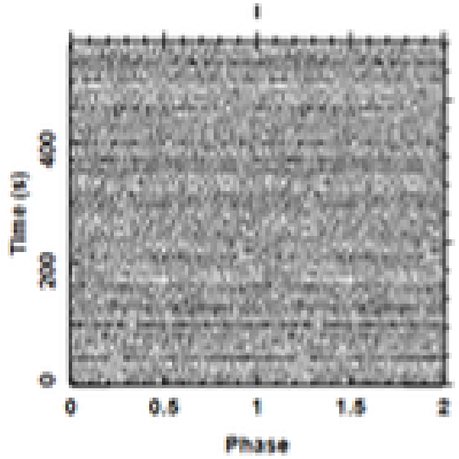
FOLDING!!!!!: ONCE A CANDIDATE PERIOD (P) AND DM ARE FOUND, THE ENTIRE TIME SERIES IS SICED INTO CHUNKS OF LENGTH P AND SUMMED (FOLDED) PHASE-BY-PHASE.

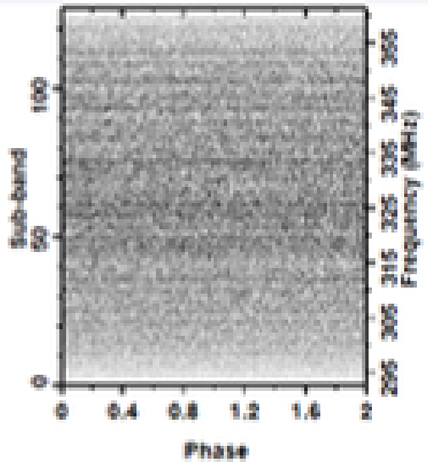
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Telescope	Arecibo	Date	2019-09-21	χ^2_{red}	1.083	max DM = 40 pc/cm^3	J2252+2455 [34.88 pc/cc] [1797.9080000000001 ms]
RA	22:51:54	DM	34.784 pc/cm^3	P(Noise)	< 0.173 (0.9σ)		J2252+24 [34 pc/cc] [1798 ms]
DEC	24:55:21	Period	0.000(14) ms	PEACE Score			J2251+24 [34.6 pc/cc] [1797.62 ms]
Discuss	Post	Data	JSON | Plot | Listen	PRESTO	Freefold		

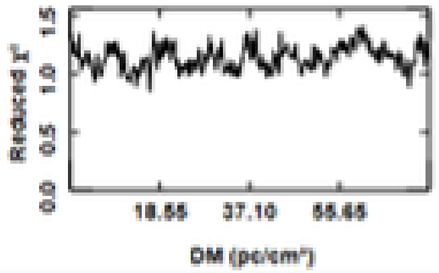
Grade

puppi_58747_J2251+24_0332_timing_masked_PSR_2251+24.pfd







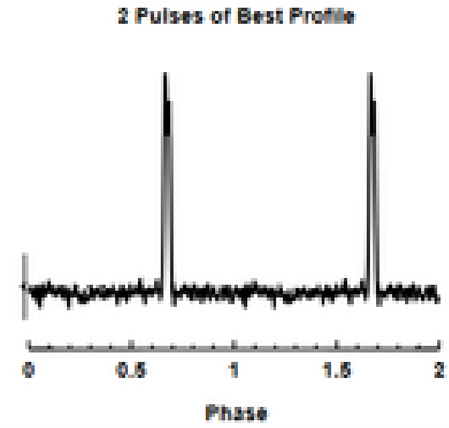


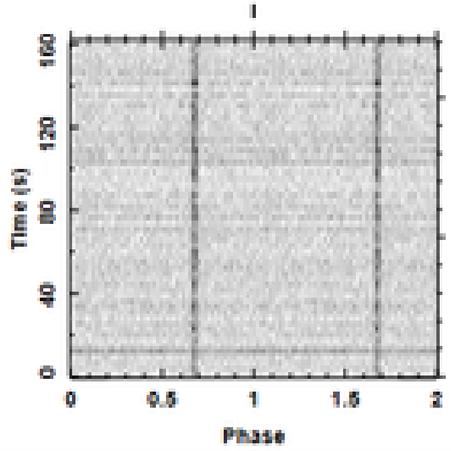
GUESS..

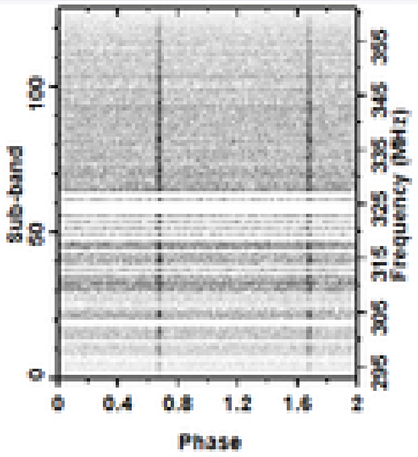
Survey	exemplars	MJD	57119.44			ion Model	Pulsar Survey Scraper
Telescope	Arecibo	Date	2015-04			c/cm^3	J1832+2758 [46 pc/cc] [631.8000000000001 ms]
RA	18:32:13	DM	47.364				J1832+27 [47.1 pc/cc] [631.7 ms]
DEC	27:45:22.001	Period	0.0000(6				
Discuss	Post	Data	JSON | Plot | Listen	PRESTO	Freefold		

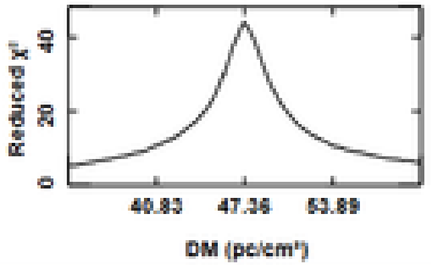
Grade

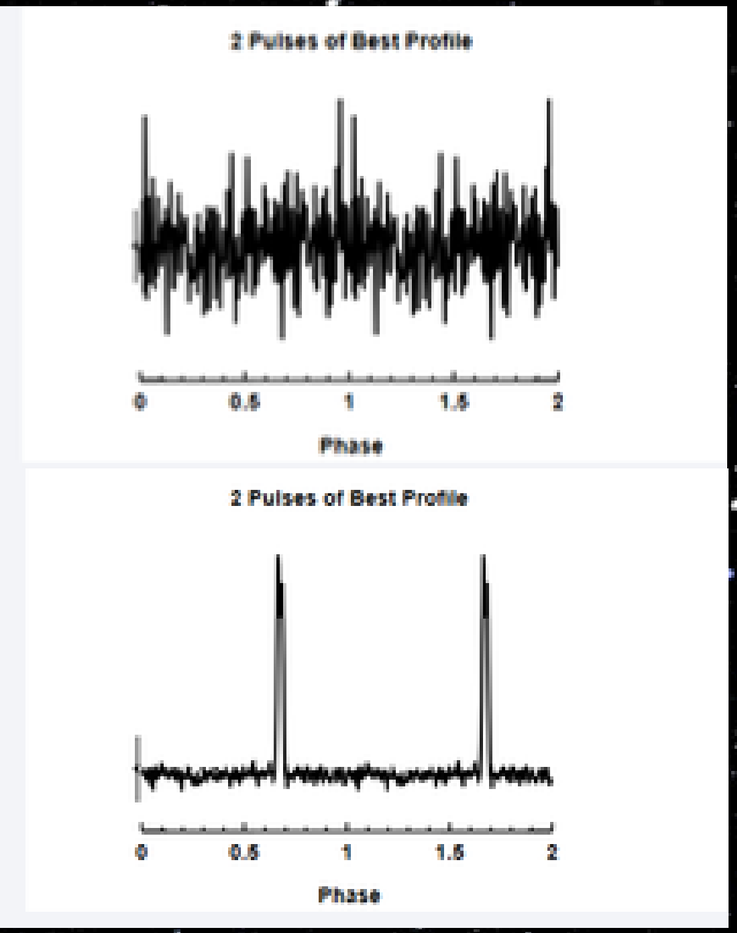
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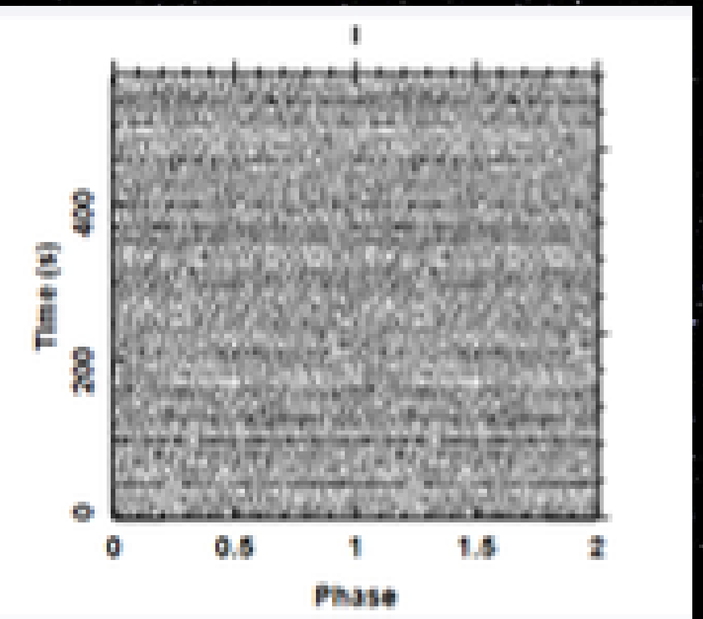


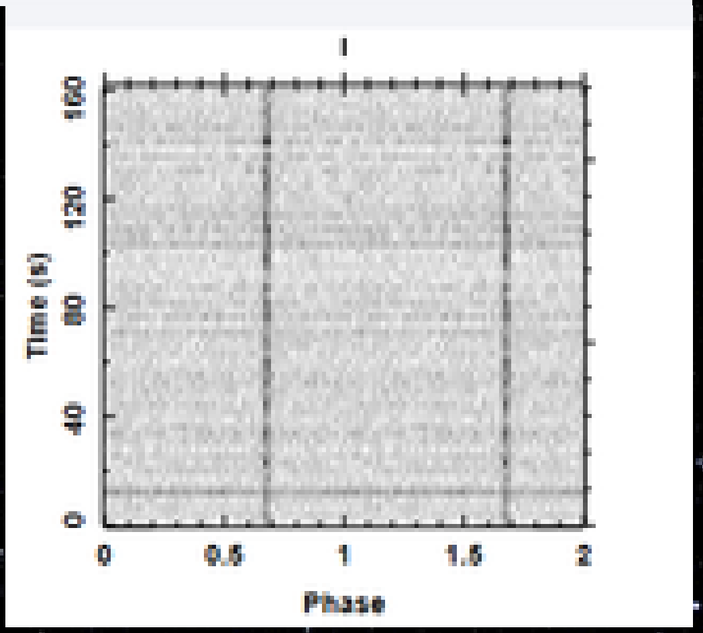




THE AVERAGE SIGNAL INTENSITY SUMMED ACROSS ALL TIME AND FREQUENCY CHANNELS OVER TWO FULL ROTATIONAL PHASES..

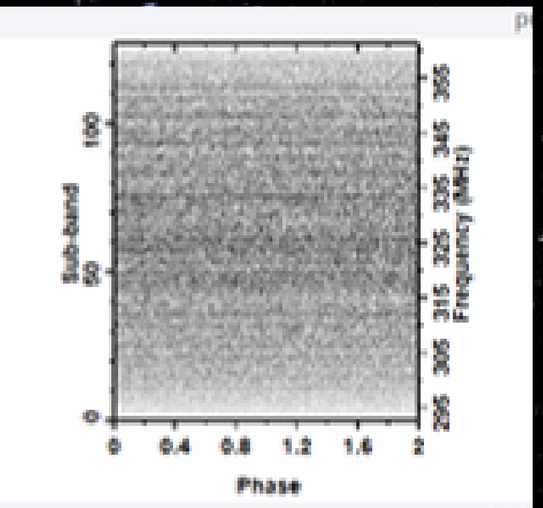
IF ITS A PULSAR.. WELL WE WOULD THINK IT PULSES. (THINK LIGHTHOUSE). WE WOULD SEE SOME PRETTY UNIFORM, LARGE, SPIKES WHEN THE PULSAR JETS ARE IN OUR DIRECTION.

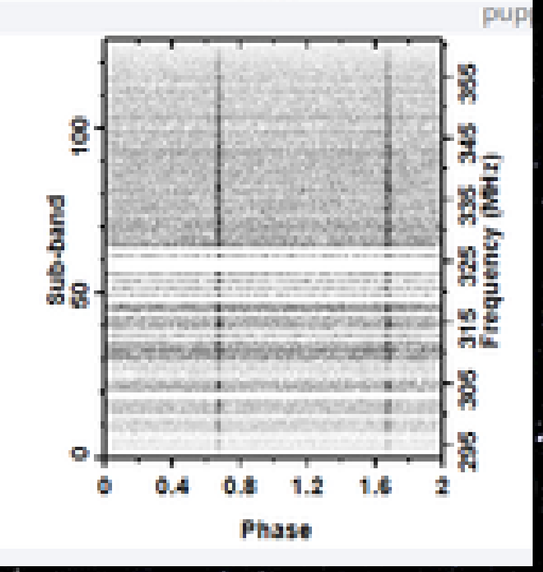




THE TOTAL OBERVATION TIME IS DIVDED INTO SUB INTERVALS ON THE Y AXIS.

EACH HORIZONTAL SLICE IS LIKE ONE FULL ROTATION... WE WOULD EXPECT A PULSE AT THE SAME PHASE EACH TIME.. STACK THOSE ON TOP OF EACH OTHER... VERTICAL LINES!



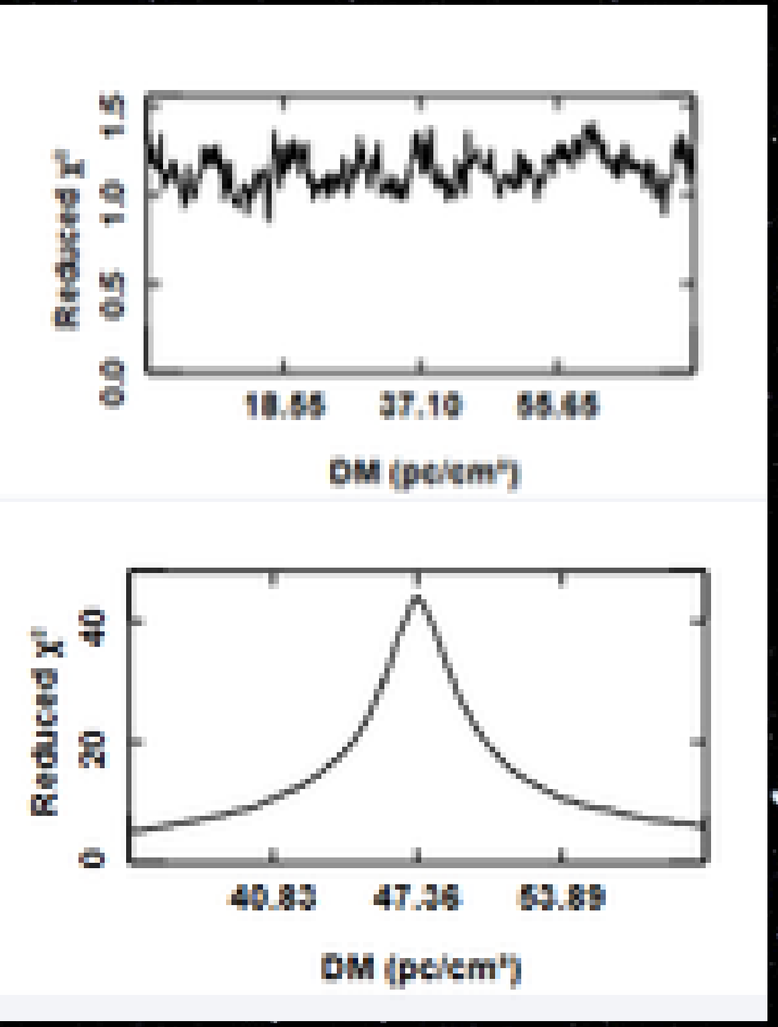


THE OBSERVING BAND (295-355 MHZ) IS SPLIT INTO SUBBANDS.

A PULSARS LIGHT OUTPUT ISNT A LASER AT A SINGLE FREQUENCY. IT EMITS A BROAD SPECTRUM OF RADIO WAVES SIMULTANEOULSY. WHEN THE BEAM SWEEPS PAST EARTH, IT FLASHES ACROSS ALL FREQUENCIES.

TO MAKE THE GRPH, WE DIVIDE THE TOTAL BANDWIDTH INTO MULTIPLE HORIZONTAL FREQUENCY SLICES RANGING FROM LOWER AT THE BOTTOM TO HIGHER AT TOP. THE PIPELINE FOLDS THE TIME DATA FOR EACH INDIVIDUAL FREQUENCY SLICE SEPARATELY TO SEE WHERE THE PULSE ARRIVES.

- THIS IS WHERE DM CAN COME IN! IF DM IS OFF, THE LOWER FREQUNCIES ARE SLOWER SO YOU GET LIKE A SLOPE. IT ALSO ELIMINATES SOME RFI CAUSE LOCAL RFI (CELL TOWERS, RADAR) ONLY AFFECT A FEW FREQUENCY CHANNELS.



THIS IS REDUCED CHI^2 VS DM. HORIZONTAL AXIS IS TRIAL DMS AND VERTICAL AXIS IS REDUCED CHI^2 WHICH MEASURES SIGNAL STNREGTH AGAINST RANDOM NOISE

DM QUANTIFIES THE TOTAL COLUMN DENSITY OF ELECTRONS BETWEEN EARTH AND THE PULSAR (INT 0-D OF N_E DL)

WE DONT KNOW THE EXACT DISANCE OR DENSITY ACROSS THE LINE OF SIGHT IN ADVANCE.. SO WE SEARCH A LOT OF DIFFERENT DM VALUES.
WE TEST EACH DM BY SHIFTING FREQUENCY CHANNELS AND SUMMING PULSE PROFILES. IF TRIAL DM WRONG, THE PULSES DONT LINE UP AND LOOK LIKE NOISE.. GIVING A LOW CHI^2. IF DM IS RIGHT, EVERY CHANNEL ALIGNS AND WE GET PROFILE VARIANCE.. SO A HIGH X^2

MAX DM. WHEN YOU POINT A TELESCOPE AT COORDINTES, THE MODEL INTEGRATS N_E ALONG THAT SIGHTLINE ALL THE WAY TO THE EDGE OF THE MILY WAY TO GIVE A BASELINE MAX.

A REALISTIC CANDIDATE (DM<= DM_MAX): IF DM IS BELOW THE MODELS LIMIT, THE SIGNALS INSIDE OUR GALAXY.

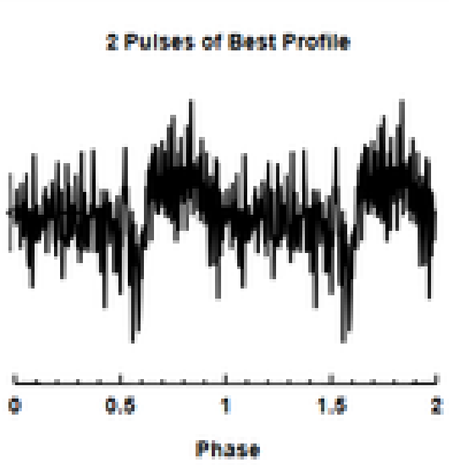
IF IT DOESNT... ITS PROBABLY RFI, AN EXTRAGLATIC OBJECT, OR WERE REALLY REALLY UNDERESTMATING LOCAL DENSE PLASMA.

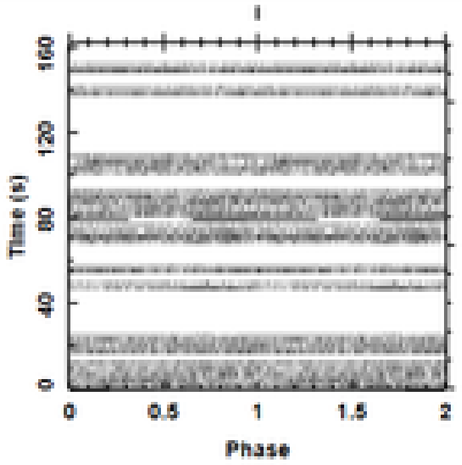
YOU TRY...

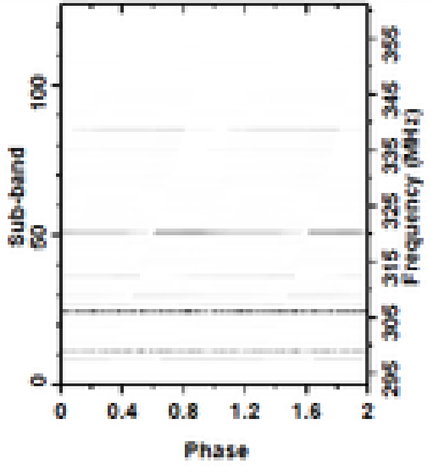
Survey	exemplars	MJD	57165.784421	Pdot	0.0(8.0)x10^{-7} s/s	Galactic Dispersion Model	Pulsar Survey Scraper
Telescope	Arecibo	Date	2015-05-23	χ^2_{red}	1.806	max DM = 212 pc/cm^3	J0608+00 [48.05 pc/cc] [1076.19 ms]
RA	06:08:48.4999	DM	48.424 pc/cm^3	P(Noise)	< 5.37e-14 (7.4σ)		
DEC	00:44:11.1998	Period	0.000(17) ms	PEACE Score			
Discuss	Post	Data	JSON | Plot | Listen	PRESTO	Prepfold		

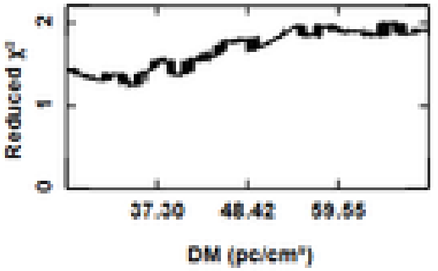
Grade

puppi_57165_J0608+00_1318_subs_timing_masked_PSR_0608+00.pfd







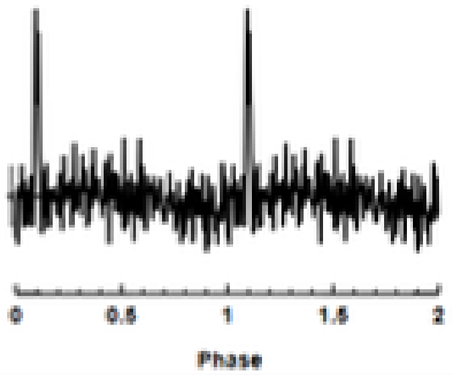


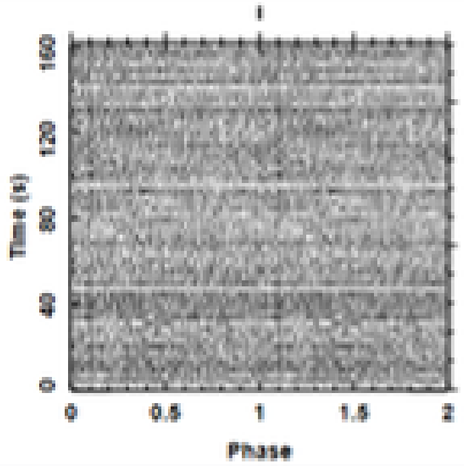
Previous [a] Skip [d]

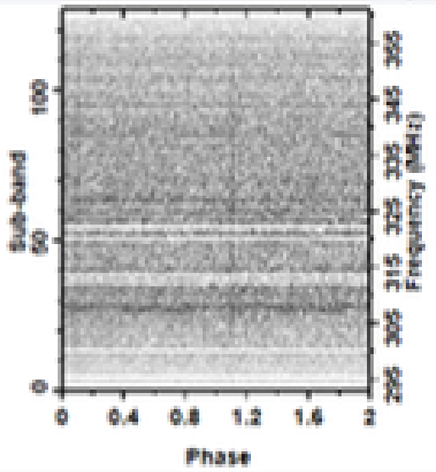
Survey	exemplars	MJD	57134.253299	Pdot	0.0(1.6)x10^{-6} s/s	Galactic Dispersion Model	Pulsar Survey Scraper
Telescope	Arecibo	Date	2015-04-22	χ^2_{red}	2.733	max DM = 40 pc/cm^3	J1628+06 [52.2 pc/cc] [1381.77 ms]
RA	16:28:53.8999	DM	51.704 pc/cm^3	P(Noise)	< 1.03e-42 (13.6σ)		
DEC	06:14:30.0004	Period	0.000(33) ms	PEACE Score			
Discuss	Post	Data	JSON | Plot | Listen	PRESTO	Prepfold		

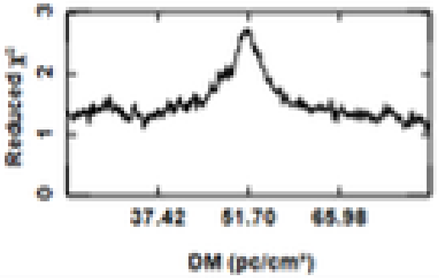
Grade

puppi_57134_J1628+06_0947_subs_timing_masked_PSR_1628+06.pfd







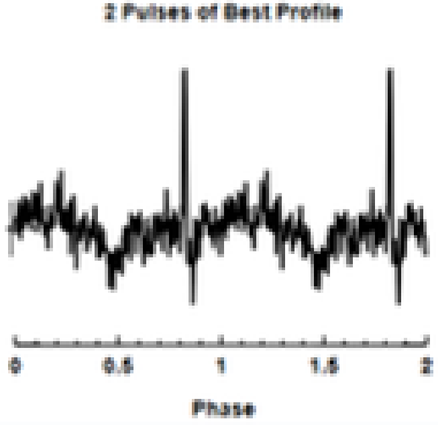


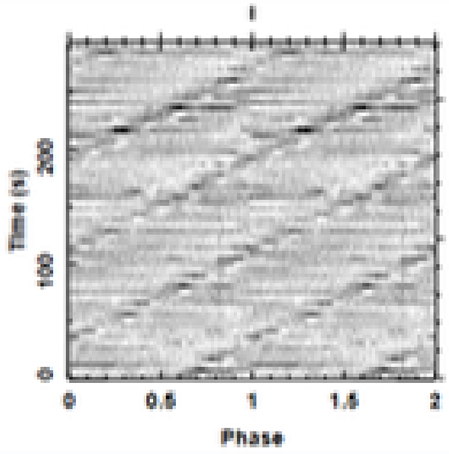
Previous [a] Skip [d]

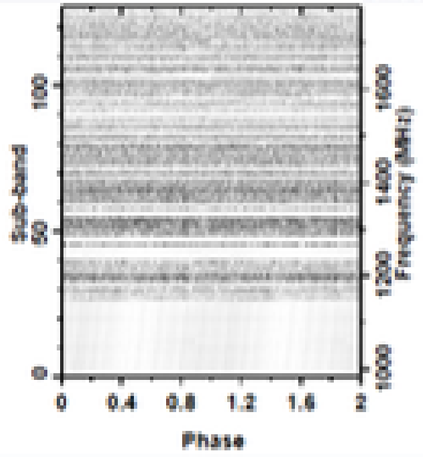
Survey	exemplars	MJD	56521.527292	Pdot	0.0(4.5)x10^{-7} s/s	Galactic Dispersion Model	Pulsar Survey Scraper
Telescope	Arecibo	Date	2013-08-17	χ^2_{red}	3.608	max DM = 212 pc/cm^3	J0608+00 [48.05 pc/cc] [1076.19 ms]
RA	06:08:49	DM	47.997 pc/cm^3	P(Noise)	< 5.96e-76 (18.4σ)		
DEC	00:44:35.9	Period	0.000(17) ms	PEACE Score			
Discuss	Post	Data	JSON | Plot | Listen	PRESTO	Prepfold		

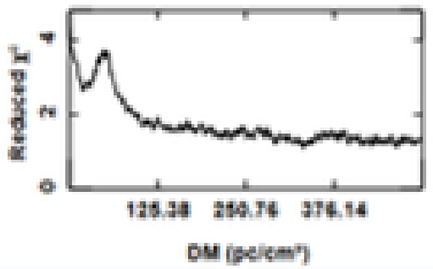
Grade

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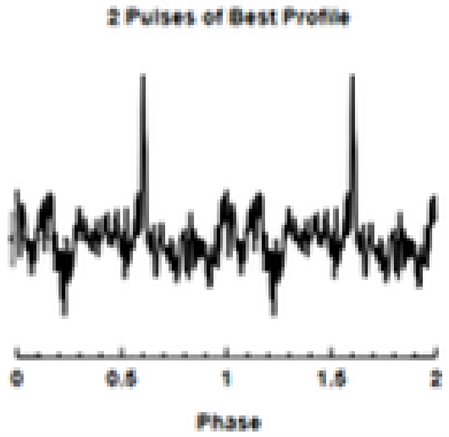


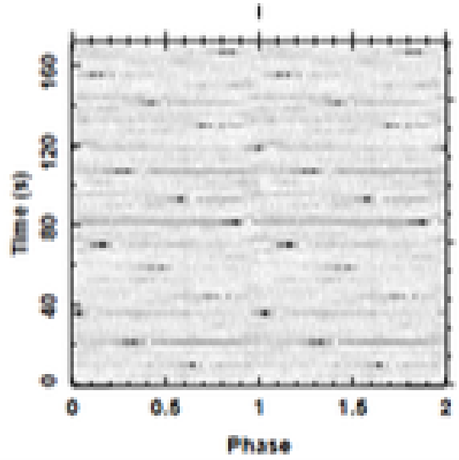
Previous [a] Skip [d]

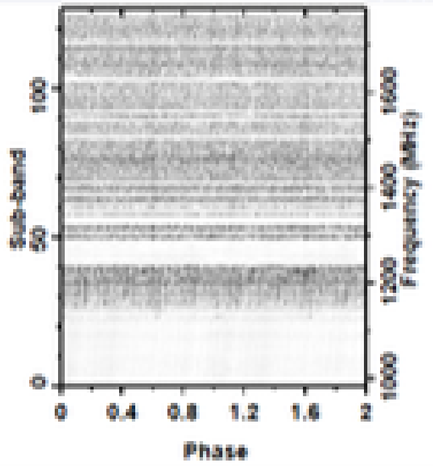
Survey	exemplars	MJD	56919.890127	Pdot	0.0(2.0)x10^{-6} s/s	Galactic Dispersion Model	Pulsar Survey Scraper
Telescope	Arecibo	Date	2014-09-19	χ^2_{red}	4.218	max DM = 40 pc/cm^3	J1628+06 [52.2 pc/cc] [1381.77 ms]
RA	16:28:53.9	DM	51.568 pc/cm^3	P(Noise)	< 3.68e-101 (21.3σ)		
DEC	06:11:59.5008	Period	0.000(44) ms	PEACE Score			
Discuss	Post	Data	JSOn | Plot | Listen	PRESTO	Prepfold		

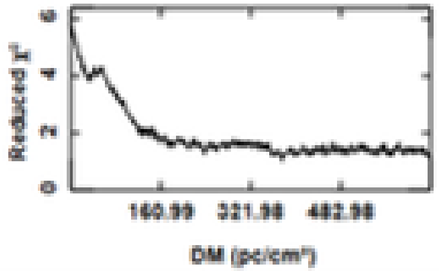
Grade

puppi_56919_J1628+06_0_0053_subs_timing_masked_PSR_1628+06.pfd









Previous [a] Skip [d]

RESEARCH 1: PULSAR SEARCH COLLABORATORY.. DO IT AND THEN JOIN OUR SCHOOL TEAM TO START GRADING.. THEN AFTER 1000 YOU CAN JOIN ONE OF THE RESEARCH PROJECTS:

RESEARCH 2: IASC

RESEARCH 3: INDIVIDUAL MENTORSHIP.. (LATER)

REMINDER FILL OUT THE OFFICER APPS!!